

VINCENT ONG

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EDUCATION

National University of Singapore, Singapore

Aug 2025 — Present

- Bachelor of Computing in Computer Science

SKILLS

Programming Languages: Python, Go, Java, TypeScript, JavaScript, SQL, R

Frameworks: React, Gin, Next.js, LangGraph, PyTorch, FastAPI, Express.js

Technologies: PostgreSQL, Docker, Git, Redis, oMLX, GCP, AWS

EXPERIENCE

AI Engineer Intern

May 2026 — July 2026

AI Singapore

Singapore

- Engineered an **agentic multimodal** document parser for CSIT orchestrating **18** open-source OCR and VLMs (GLM-4.6V, Qwen3-VL) through layout-aware, region-level tool dispatching, yielding **97% extraction fidelity**.
- Designed a **self-improving evaluation harness** pairing cross-family **VLM-as-a-Judge** scoring with deterministic guardrails, stateful agent memory and robust regression test coverage, cutting CER and WER by **~27%**.
- Deployed on-prem, serving quantized VLMs on Apple Silicon via **MLX/oMLX** instrumented with **OpenTelemetry** and **Arize Phoenix** tracing. Rigorously **A/B tested** agentic variations against deterministic baselines.

Software Engineering Intern

Mar 2025 — Jun 2025

Sonzai Labs

Singapore

- Shipped a high-concurrency clan system for the startup's flagship game, Telemafia, serving **22 RESTful APIs** in **Go**, with two-tier **Redis** caching and atomic **ScyllaDB CQL** counters, eliminating write-lock contention.
- Engineered **responsive, high-performance** gameplay UI/UX for global users with **React, TypeScript, Next.js, i18n** for internationalization and rapid bug resolution on a **CI/CD** workflow with **Vitest** and **Playwright E2E** tests.
- Led frontend development** for the secure launch and minting of **500 NFTs** by integrating **smart contracts** with Web3/TON wallet authentication and configuring **Cloudflare WAF** to mitigate DDoS and XSS attacks.
- Developed a real-time leaderboard system with **Python** pipelines on **Windmill Orchestrator** streaming **Apache Pulsar** (Pub/Sub) events into **ScyllaDB** for **automated data reconciliation**, driving a **10x throughput gain**.

SAF Infantry 3rd Sergeant

Mar 2023 — Jan 2025

Singapore Armed Forces

Singapore

- Demonstrated **reliability** and **trustworthiness** in high-stakes environments, overseeing firearms **accountability** and taking **ownership** of the safety and operational readiness of recruits while **leading** as a section **commander**.

PROJECTS

ThreadTalk | Go, Gin, Next.js, TypeScript, React, AWS, Docker, PostgreSQL

Dec 2025

- Architected a scalable, Reddit-inspired forum application by integrating **Go (Gin)** RESTful APIs with a Next.js, TypeScript and React frontend, backed by a **PostgreSQL** relational database.
- Optimized data retrieval for **deep discussion threads** using recursive hierarchical comment nesting and infinite scroll with **concurrent processing** for speed.
- Automated secure production deployments by containerizing the serverless backend with **Docker** for **AWS Lambda**, establishing a **CI/CD** pipeline via **GitHub Actions** and **OIDC** for keyless **Amazon ECR** integration.
- Hardened application security and data integrity using **AWS VPC** isolation, **IAM** least-privilege policies and **JWT** authentication while automating **RDS** schema migrations with **Go**.

Bastion, Sup x OpenAI Hackathon | Cloudflare Workers, D1, KV, Workers AI, React

June 2026

- Shipped an edge-native **threat detection** layer that **intercepts and denies** malicious tools, skills and URLs before AI agents execute them, fail closed, anchoring every verdict in an independently verifiable **SHA-256 hash chain**.

Baitrage, AI Engineer Hackathon | Vercel AI SDK, Zod, Web Speech API, Chokidar

May 2026

- Built a **multimodal** frustration detector scoring voice, facial affect and screen context with **Gemini Flash**, triggering a **RAG agent swarm** over a live SSE symbol map to synthesize codebase-aware prompts that break error loops.

NUS Fintech Summit Hackathon | XRP Ledger SDK, Python, FastAPI

Jan 2026

- Architected **blockchain rails** on **XRP Ledger**. **Eliminated 100% pre-funding requirements** by deploying **XRPL escrows** and **agent-driven** credit underwriting, replacing **collateral** locks with conditional **settlement**.

Visente | Entrepreneurship

Nov 2022 — Mar 2023

- Generated approximately **\$100 profit** by launching a niche fragrance decanting venture, **selling 40+ bottles** by leveraging consumer demand for **affordable access to diverse scents**.
- Identified high-margin opportunities by executing strategic market research on **competitor pricing** and **consumer trends**, using **fragrance profiling** to optimize **cost efficiency** and production selection.

CERTIFICATIONS

Google Cloud Professional Machine Learning Engineer Certification

Jun 2025

Google • Expires: Jun 2027